INCH-POUND

MIL-I-7088B
30 JANUARY 1993
SUPERSEDING
MIL-I-7088A
18 JUNE 1979

MILITARY SPECIFICATION

INDICATOR, PRESSURE, FUEL, 0-1,000 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hermetically sealed, remote indicating, synchro-style fuel pressure indicator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unliss otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-5396	Transmitter, Pressure, Fuel, 0-1. 000 PSI
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-2073/1A	DOD Material Procedures For Development & Application Of Packaging Requirements
MS28005	Transmitter, Pressure, Synchro, Aircraft
MS28010	Indicator, Pressure synchro, Single, 2-Inch Size
MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AF Drawing 8988994 Luminescent Material, Fluorescent

(Copies of Air Force drawings are available from OC-ALC/TILDOS, Tinker AFB, OK 73145-5990.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing and Materials (ASTM)
ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to : ASTM, 1916 Race St, Philidelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated

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detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in specification MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.1.1 General specification. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design. The indicator shall be designed for use with an MS28005-4 pressure transmitter in accordance with specification MIL-T-5396.

3.3 Construction.

3.3.1 Case. The case shall be hermetically sealed and the case dimensions shall conform to standard MS28010.

3.3.1.1 Sealed cap. A separate nonhermetically sealed cap may be used on the back of the case for the purpose of installing the electrical receptacles. The cap shall be made of nonferrous, low-density metal.

3.3.1.2 Cover glass. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.3.2 Pointer. The pointer shall be in accordance with MS33585-8, except that the length of the pointer shall be such that the tip will extend into the scale a distance equal to 1/3 to 2/3 the length of the shortest graduation.

3.3.2.1 Pointer finish. The shaded portion of the pointer shall be finished with fluorescent-luminescent material.

3.3.3 Dial. The dial shall be as shown on figure 1. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The graduation for 300 psi shall be located on the horizontal center line at the left side of the dial.

3.3.3.1 Dial markings.

3.3.3.1.1 Fluorescent-luminescent markings. The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

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	Height or Length Inch +1/64	Width of Line or Graduation Inch +0.005
Numerals 0, 1, 2, 4, 5, 8, and 10	5/32	0.025
Graduations at 0, 100, 200, 300, 400, 600, 700, 800, 900, and 1,000	3/16	0.031
Graduation at 500	9/64	0.031
20 psi graduations from 0 to 100	3/32	0.020
50 psi graduations	3/32	0.020
Lettering "FUEL PRESS"	1/8	0.025
Lettering "PSI X 100"	3/32	0.020

3.3.3.1.2 Durable dull black markings. The markings for the applicable MS part number shall be permanently and legibly marked on the dial in letters 1/16 inch high; location is optional. These markings and all other markings not otherwise specified shall be furnished in durable dull black.

3.3.4 Weight. The weight of the indicator shall not exceed 0.60 pound.

3.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.5 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective materal, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Fuel pressure. The fuel pressure indicator shall be subjected to inspection and tests in accordance with specification MIL-I-7057.

4.3 Test tolerances. The indicator shall be within the tolerances specified in Tables I and II of this specification when tested in accordance with specification MIL-I-7057.

TABLE I

Pressure Indicator

PSI	Tolerance + psi
0	3.0
60	3.0
100	3.0
200	3.0
300	3.0
400	3.0
500	3.0
600	3.0
700	3.0
800	3.0
900	3.0
1,000	3.0

TABLE II

Pressure Indicator

Test	Tolerance + psi
Position error at 500 psi	5.0
Friction	
Room temperature	5.0
High temperature	5.0
Low temperature	8.0
Vibration	
Point oscillation	5.0
Pointer variation	5.0
Scale error	
Low temperature	5.0
High temperature	5.0

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4.4 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116, to ensure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C, or Industrial IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-1 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method ICI of MIL-P-116. Each indicator shall be placed in a PPP-B-636 fiberboard container, weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to ensure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of the indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be Level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare, and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

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5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed IAW ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fuel pressure indicator covered by this specification is intended for use in conjunction with a suitable transmitter in indicating remotely the pressure in the fuel systems of aircraft engines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 General. The instruction contained in section 6 of specification MIL-I-7057 are applicable to and form a part of this specification.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/TICLA, Tinker AFB OK, 73145-5990, and information pertaining to qualification of products may be obtained from that activity.

6.5 Subject term (key word) listing.

Hermetically sealed
Synchro-style fuel pressure indicator

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-7088B	2. DOCUMENT DATE *(YYMMDD)* 93/01/30

3. DOCUMENT TITLE
INDICATOR, PRESSURE, FUEL, 0-1,000 PSI

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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